Topological Protection of Vacuum Energy Sign: A Testable Prediction from K-Theory for Dirichlet–Neumann Casimir Scaling

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The Casimir energy of a one-dimensional cavity changes sign when the boundary conditions are changed from Dirichlet–Dirichlet (DD) to Dirichlet–Neumann (DN): $E_{DD} \propto \zeta(-1) = -1/12 < 0$ (attractive) versus $E_{DN} \propto \zeta_{\neg 2}(-1) = +1/12 > 0$ (repulsive). This is well established. We observe that the localization $\mathbb{Z} \to \mathbb{Z}[1/2]$ underlying the DN configuration induces a change in algebraic K-theory, $K_1(\mathbb{Z}) = \mathbb{Z}/2 \to K_1(\mathbb{Z}[1/2]) = \mathbb{Z}/2 \times \mathbb{Z}$, introducing a topological winding number that protects the sign of the vacuum energy. If this protection is physical, the standard $1/a^4$ decay of the Casimir energy density must exhibit a *deviation* at a critical separation a_c where the topological transition occurs: the density should plateau rather than continue to decay. We propose a minimal experiment—precision measurement of the DN Casimir force scaling law—that can confirm or rule out this prediction. The required apparatus (Si-MEMS or circuit QED) exists.

THE SIGN FLIP

Consider a massless scalar field in a one-dimensional cavity of length a. With Dirichlet conditions at both ends (DD, "$\lambda/2$"), the mode frequencies are $\omega_n = n\pi c/a$, $n = 1, 2, 3, \ldots$, and the regularized vacuum energy is

$$E_{DD} = \frac{\pi\hbar c}{2a}\,\zeta(-1) = -\frac{\pi\hbar c}{24a} < 0. \tag{1}$$

With Dirichlet at one end and Neumann at the other (DN, "$\lambda/4$"), the modes are $\omega_n = (2n-1)\pi c/(2a)$, $n = 1, 2, 3, \ldots$, giving

$$E_{DN} = \frac{\pi\hbar c}{4a}\sum_{n=1}^{\infty}(2n-1) = \frac{\pi\hbar c}{4a}\,\zeta_{\neg 2}(-1) = +\frac{\pi\hbar c}{48a} > 0. \tag{2}$$

Here $\zeta_{\neg 2}(s) \equiv (1-2^{-s})\zeta(s) = \sum_{n\ \mathrm{odd}} n^{-s}$ is the Riemann zeta function with the p=2 Euler factor removed. The *sign flip* $E_{DD} < 0 \to E_{DN} > 0$ is a standard result, confirmed by multiple regularization methods [1–4].

In three spatial dimensions, the analogous quantities involve $\zeta(-3) = 1/120$ and $\zeta_{\neg 2}(-3) = -7/120$, giving Casimir energy densities

$$\rho_{DD} = -\frac{\pi^2\hbar c}{720\,a^4}, \qquad \rho_{DN} = +\frac{7\pi^2\hbar c}{720\,a^4}. \tag{3}$$

THE K-THEORETIC OBSERVATION

The passage from DD to DN corresponds algebraically to the localization $\mathbb{Z} \to \mathbb{Z}[1/2]$: one removes all integers divisible by 2 from the mode sum, retaining only the odd modes. This localization induces a change in algebraic K-theory [5]:

$$K_1(\mathbb{Z}) = \mathbb{Z}/2 \;\longrightarrow\; K_1(\mathbb{Z}[1/2]) = \mathbb{Z}/2 \times \mathbb{Z}. \tag{4}$$

The emergent $\mathbb{Z}$ factor is a *winding number*. In condensed-matter physics, an analogous winding number (the SSH invariant [6]) topologically protects edge states: a system with winding number $w \neq 0$ cannot be continuously deformed to $w = 0$ without closing the gap.

We conjecture that the $\mathbb{Z}$ winding number in (4) *topologically protects the sign* of the vacuum energy in the DN configuration: once the system is in the $E > 0$ (repulsive) phase, it cannot be continuously deformed back to $E < 0$ (attractive) without a discrete topological transition.

THE PREDICTION: SCALING DEVIATION

In standard quantum field theory, the Casimir energy density $\rho \propto 1/a^4$ for both DD and DN boundary conditions. As the plate separation a increases, ρ decays to zero. The sign difference between DD and DN is maintained at every a, but the *magnitude* vanishes as $a \to \infty$.

If the sign of the vacuum energy is topologically protected, we predict a qualitatively different behavior: there exists a critical separation a_c such that for $a > a_c$, the energy density ρ_{DN} *ceases to decay* as $1/a^4$ and instead plateaus at a constant value determined by the topological invariant.

Regime	Standard QED	Topological
$a < a_c$	$\rho \propto 1/a^4$	$\rho \propto 1/a^4$
$a > a_c$	$\rho \propto 1/a^4$	$\rho \to$ const
Scaling exponent	-4 everywhere	$-4 \to 0$ at a_c

The critical separation a_c is not predicted by our analysis. It could range from nanometers (if set by the lattice of Spec($\mathbb{Z}$)) to macroscopic scales (if set by cosmological parameters). The prediction is the *existence* of the deviation, not its location.

ANALOGY: BULK–EDGE CORRESPONDENCE

In topological insulators, the *bulk–edge correspondence* guarantees that a material with nontrivial bulk topological invariant must have gapless edge states, regardless of surface details. The edge states are "protected" by the bulk.

We propose an analogous *bulk–boundary correspondence* for the vacuum: the sign of the Casimir energy (a boundary effect) is protected by the K-theoretic invariant of the bulk vacuum ($K_1(\mathbb{Z}[1/2])$). Just as the conductance of a topological edge state is quantized and insensitive to disorder, the sign of E_{DN} should be insensitive to the details of the boundary, depending only on the topological class.

This would manifest experimentally as a deviation from the $1/a^4$ scaling: the "boundary" effect (Casimir energy) would acquire a "bulk" contribution (constant energy density) once the topological protection is established.

EXPERIMENTAL TEST

Minimal experiment: Measure the DN Casimir force as a function of plate separation a over at least two decades (e.g., $a = 50\,\text{nm}$ to $5\,\mu\text{m}$) and fit the power law $F \propto a^{-\alpha}$.

- Standard QED predicts $\alpha = 5$ (force $\propto -dE/da \propto a^{-5}$ in 3D) for all a.
- The topological prediction is $\alpha < 5$ for $a > a_c$, ultimately approaching $\alpha = 1$ (constant density $\to$ force $\propto 1/a$... actually $F = -d(\rho V)/da$ if ρ constant then $F = -\rho A$ independent of a, i.e., $\alpha \to 0$).

Any confirmed deviation of α from 5 in the DN configuration, not present in the DD configuration, would be evidence for topological protection of the vacuum energy sign.

Platforms:

- **Si-MEMS** (Princeton-type [7]): Geometric Casimir force measurement at 50–500 nm. DN boundary conditions achievable via T-shaped or open-ended structures.
- **Circuit QED**: $\lambda/4$ vs. $\lambda/2$ superconducting resonator with tunable end impedance Z. Sweep Z from 0 (Dirichlet) to ∞ (Neumann) and measure Lamb shift scaling.

DISCUSSION

We emphasize that this paper makes a *single, falsifiable prediction*: the DN Casimir force deviates from $1/a^5$ scaling at some separation a_c. If no deviation is found across the accessible range, the topological protection conjecture is excluded for a_c within that range.

The prediction is independent of the broader "Spec($\mathbb{Z}$) as spacetime" program. It relies only on: (i) the algebraic K-theory computation (4), which is a theorem; (ii) the physical identification of K_1 winding with vacuum energy sign, which is the conjecture.

If confirmed, the implications extend far beyond the Casimir effect: a topologically protected constant vacuum energy density would constitute a *controllable dark energy*, with applications to spacetime engineering.

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